Systemic Problem Analysis

IIIT Hyderabad Breakfast Mess System

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Overview

Students at IIIT Hyderabad register for breakfast in large numbers and most of those registrations go uneaten. Kadamba took 32,162 breakfast registrations in April 2026 and scanned 11,941 at the counter: 37.1% turnout, 62.9% no-shows. Yuktāhār's September register is legible on thirteen days, and breakfast turnout on those days ran from 17.7 to 40.2%, about 28% on average, while lunch in the same hall ran from 56 to 89%. Both figures are confirmed from the halls' own records. Where lunch is recorded too, the gap belongs to breakfast.

The analysis works in layers. An event is one thing that happened. A pattern is what keeps happening across events. A structure is the rule or design that produces the pattern, and a mental model is the belief that keeps the structure standing. Read from the bottom, a belief shapes a rule, the rule produces a habit, and the habit shows up as an incident you can point to.

Claims carry four labels. Confirmed means read off a record, seen on site by a team member, or stated clearly by a named source. Single-sourced means one account. What the Kadamba operator told a team member who heard it directly is single-sourced, and the figure is not in doubt, though it remains the operator's own account of its practice. The Bakul manager's account reaches us through a team member's synthesis of the recordings, and no Bakul record of any kind has been seen, so every Bakul number is the manager's statement or estimate. Candidate is a reasonable reading not yet checked, and assumed is a systems judgement offered without proof.

Two findings carry the analysis. First, the institution has already run a version of breakfast where billing follows who shows up. During a gas shortage, and on holidays when students leave, registration stopped being compulsory, and a student who neither registered nor came was not charged. The model was switched off both times when the trigger passed, and nobody seems to have asked whether any of it should stay. Second, the gap is measured twice and reaches no one who can change the rules. The central dining system logs every scan against every registration. Yuktāhār's operator, ABC Hospitality Services, records per meal and per item how many registered, how many ate,

what was cooked, what was left and when each item ran out. Meanwhile three kitchens plan three ways, and in the two halls with records the breakfast gap is about the same size under opposite philosophies. The kitchen is rich in data and poor in authority. The offices that own registration and billing hold the authority and not the data.

1. Events, Patterns, Structures and Mental Models

Seven areas were traced from event up to belief. Each keeps its own layers, because the point is to see how many different rules and beliefs sit under one visible symptom.

The seven chains at a glance

One visible event in each chain, traced down to the belief that holds it in place

	Event	Pattern	Structure	Mental model	Root cause
1 Registered, not showing up	Sunday 6 Sept: 267 registered at Yuktāhār, 50 ate	Flat 1,070 a day at Kadamba; turnout 27 to 42%	Automatic default; five cancellations; billed at booking	"We have no opportunity to nullify it"	Billing tracks registration, not attendance
2 Food runs out at the close	Idli and puri gone by 9:20; Kadamba last scan 10:07	26.5% of scans in 9:15 to 9:30; Sunday latest	First batch fixed before any signal; top-up only upward	"The highest flow is in the last 10 minutes"	Late one-way top-up against a crest
3 Complaints ignored	Cap tightened Dec 2024; water complaints unheard in 2023	A loud problem is fixed, a quiet one is not	Menu, kitchen, calendar, billing with different owners; oversight on safety and taste only	Consultation is optional; workload is the reason	Nobody owns the whole picture; action follows loudness
4 A fix switched off	Gas shortage and holidays: no compulsory registration	Worked every time; reverted every time	No review exists; the cap has been moved before	An automatic count beats matching demand	The institution solved it twice and did not keep it
5 A fix and its shadow	Fortnightly menu rotation replaces a fixed semester menu	Fatigue fell; turnout moves with the item	Student Council, Parliament, Committee path worked	Menu variety is worth a policy change	Loud problems get fixed; visibility also enables avoidance
6 Not a problem at all	A student fasts on purpose	Late sleep, Sundays lowest, ceiling on variation	Default registers the absent and fasting as no-shows	Sleep outranks a booked breakfast	A real slice, too small to explain the gap
7 Cook to a guess, absorb the rest	Site head sees 18.7%, still says about 50%	70% batch is about 1.9 times turnout; surplus routes	Three ratio regimes; records stay in the kitchen	"We are paid for the registration"; "waste is money"	Cooked food cannot be shrunk; the booking is upstream

Figure 1. The seven chains at a glance: from a visible event down to the belief that holds it in place.

Chain 1: Students register and don't show up

Layer	What the evidence shows
Events	A student books Kadamba veg breakfast at Rs. 48 every day and skips most days; they cancel about five times a month and have never used Skip Meal because "it doesn't give any money back". Another books the whole month because Kadamba "fills up", and resells only when a slot fetches a good price. On Sunday 6 September 267 students were registered at Yuktāhār and 50 ate (18.7%). On Sunday 5 April 1,077 were registered at Kadamba and 273 scanned in (25.3%); only 20.5% of the vegetarian registrations that morning were used (confirmed).
Patterns	Registration beats attendance every day, far more at breakfast than lunch. Kadamba's April turnout was 37.1% (32.4% veg, 48.5% non-veg); Yuktāhār's September breakfast averaged about 28%, below the 35 to 45% band the CFS Chair gave. Registrations are flat across the week, about 1,070 a day at Kadamba with Sunday included, while turnout ran from 42.1% on Tuesdays to 27.3% on Sundays (confirmed). Flat registration is what a default looks like in the records. Bakul's manager estimates at most about 60% turn up (single-sourced, no records); if Bakul resembles the two recorded halls, that is an upper bound (candidate). Nobody interviewed has ever got anything from Skip Meal, and four of six use Mess Cell regularly.

Layer	What the evidence shows
Structures	Billing counts registrations, not eaters. Every student is registered automatically, on campus or not, holidays included, and the only exit is five cancellations per meal type per month, each at least four days ahead; beyond them the charge cannot be removed (confirmed from the dining poster and a team member's own experience as a registrant). The four-day close applies to registration. Kitchens fix quantities the day before. Walking in unregistered costs about 70% more (Rs. 81 against Rs. 48 for vegetarian), and walk-ins stay small: none to twelve a morning at Yuktāhār. Mess Cell lets a student sell a slot they will not use. Skip Meal tells the kitchen and pays the student nothing, and whether it changes what Kadamba cooks is open: the institution's account says it reaches the kitchen, while the Kadamba operator says it cooks on registrations alone (single-sourced).
Mental models	The mess office treats registration totals as a good enough stand-in for demand, and keeps billing that way although a working alternative has been run. It is the strongest belief in the analysis because it survives contrary evidence. Students hold that the booking is not theirs to manage: ”We don't have any opportunity to nullify it.” For one student, resale pays and Skip Meal does not, so Skip Meal is pointless. Kadamba's operator treats the count as the obligation it is paid for and cooks ”because it is a responsibility and they are getting paid for it”; Yuktāhār's treats waste as a cost and a profit lever (both single-sourced). Neither has a route to the registration rule.

The Kadamba manager also believes a friend often eats in an absent registrant's place (single-sourced). The scan export carries no student ID, so the records cannot confirm it.

Root cause. Billing tracks registration on a booking the student never actively makes. The counterfactual has happened: the CFS Chair says the no-show problem eased when billing followed attendance during the shortage and holidays. No attendance record from those periods has been seen, so the size of the effect is unmeasured. The same gap, 37% at Kadamba and about 28% at Yuktāhār, appears under opposite kitchen philosophies, which puts the cause upstream of the kitchen.

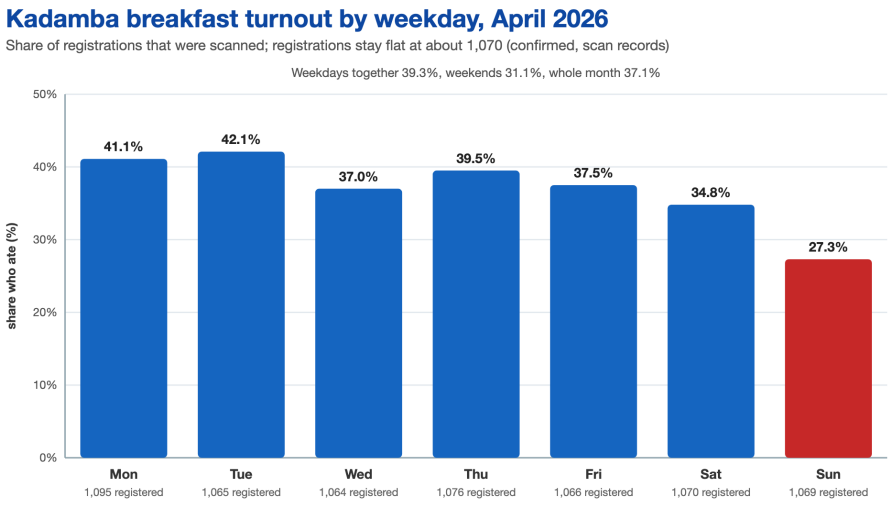


Figure 2. Kadamba breakfast turnout by weekday against flat registrations, April 2026.

Chain 2: Food runs out before the mess closes

Layer	What the evidence shows
Events	Around 9:20 most mornings idli, puri and bhatura are gone at Kadamba, fruit is not restocked, and staff promise more before the 9:30 close that usually does not come. On Thursday 3 September Yuktāhār re-cooked a kilogram of semiya upma at 9:00; that morning's production sheet shows 77 of 278 registered eating (28%) and the attendance register 110 (40%), two confirmed records that disagree. On Friday 3 April Kadamba's last breakfast scan was at 10:07.

Layer	What the evidence shows
Patterns	Crowding builds to one crest. At Kadamba in April 26.5% of scans fell between 9:15 and 9:30 and 13.5% between 9:00 and 9:15, with a small burst of 9.6% at 7:30; the rate reaches about nine a minute at 9:29 (confirmed). The Yuktāhār site head describes the same closing rush. Kadamba's operator puts it at about 100 diners early, 100 in the middle and 150 in the last ten minutes, which matches the shape. Bakul's manager describes a different curve, a main rush at 8:00 to 8:30 and 10 to 15 in the last ten minutes (single-sourced, no scans), so curves differ by hall. The crest is latest on Sundays, when 45.9% of Kadamba's scans come after 9:15 against about 32% on Tuesdays, and with no classes that day it is partly a sleep pattern (candidate). Every April day has scans at or after 9:30, 9.2% in all, about 37 a day; latecomers use the back door and ”mess can't say no to them” (single-sourced). One morning therefore holds a run-out at the counter and surplus in the pot.
Structures	The first batch is fixed before any turnout signal. Kadamba cooks 70% of registrations, veg and non-veg planned separately; Bakul 70 to 80% by item from its manager's weekly review; Yuktāhār estimates each meal from the matching week in its books (conversion regimes are compared in section 2). Top-up exists everywhere and only moves upward. Kadamba's supervisor tells the kitchen how much more to cook against gram standards, ready in 10 to 15 minutes. Bakul's manager checks about every thirty minutes and phones the Hafizpet kitchen; the drive alone is about 20 minutes early and 40 later, the total lead time is unstated, and he aims to hold food for about 50 at 9:20 against 10 to 15 late arrivals, a reserve that a top-up too slow for the crest would explain (candidate). Yuktāhār checks every thirty minutes from about 8:45 and projects the rest of the meal from the first half hour, which carried about 15% of Kadamba's April scans against 26.5% in the last fifteen minutes. Before service every Kadamba meal passes cooking temperature, a 72-hour sample, a wrapped display plate, serving temperature and a tasting with two Prism staff (plate and tasting observed, confirmed), so the batch must exist, cooked and checked, before any diner. Cooked food lasts three to four hours and a cooked batch cannot shrink. Bakul's labour-heavy items may start around 3 AM for a 7 AM arrival and sit in hot-water holding, which keeps temperature and softens aloo paratha (single-sourced). Portions are set per item: every Kadamba dish has a gram standard, and the items diners take above it (bonda, puri, vada, idli) are the ones students say run out first (candidate). Kadamba buys dry goods a week ahead, some vegetables two to three days ahead and perishables the day before.
Mental models	The Yuktāhār site head carries a breakfast figure of ”about 50%” (”at most 50%” on another occasion) against a September register of 18 to 40%, about 28 on average (confirmed); the operator with the best records carries a headline above them. Kadamba has ”no idea” of forecasting beyond ratio and gram standards: ”it was wasted many times”. The Bakul manager believes Kadamba sees 90 to 95% turnout, which Kadamba's own records contradict for breakfast. ”Student fluctuation is very high”, so Yuktāhār cooks once. Low Sundays and exam weeks ”cannot be helped”. Bakul's rule is ”there should not be a shortage of food, even if some wastage occurs”. Kadamba's operator reads the crest correctly: ”the highest flow is in the last 10 minutes.”

Root cause. The kitchen can react to a same-day spike, but only upward and late: ten to fifteen minutes on site at Kadamba against a crest of nine scans a minute, and a call, a cook and a drive at Bakul. Students' reports that the restock never arrives fit a top-up that lands after the crest. Yuktāhār's first-half-hour projection is the one forward correction and it samples the calm opening. The same morning therefore produces a shortage at the counter and a surplus in the pot.

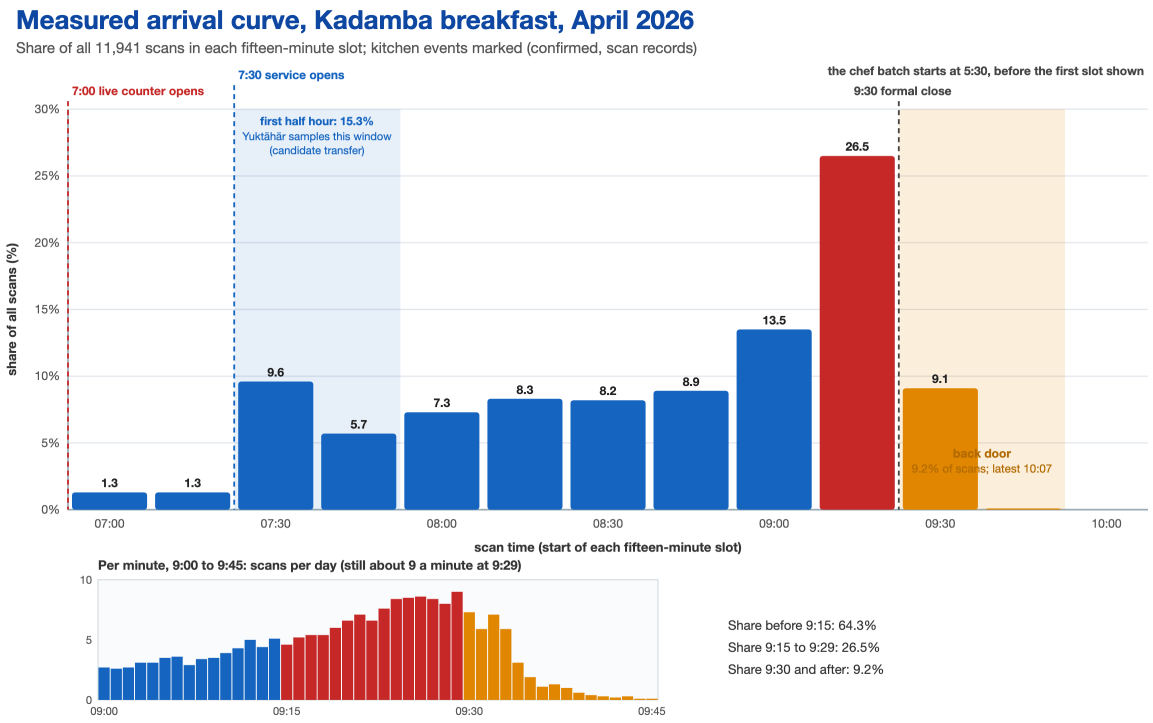


Figure 3. *Measured arrival curve, Kadamba breakfast, April 2026.*

Chain 3: Complaints get ignored until something forces a response

Layer	What the evidence shows
Events	In November 2024 a frog was reported in Kadamba's chicken biryani and non-veg meals were pulled without an official reason. In December 2024 the cancellation cap was tightened after a Kadamba renovation added to Mess Office workload, with no record of committee sign-off; a Ping survey found most respondents unhappy, and Parliament representatives met the Mess Office twice before it went ahead anyway. In 2023 complaints about drinking water went unanswered for months until more than 38 confirmed typhoid cases forced a response. In August 2025 Yuktāhār staff feared blame when waste weighing began (”they are catching us”), and by December they reported waste had fallen (confirmed). In September 2026 a Kadamba student complained that dinner sambar had been dropped ”while they are taking more money from us”, and the operator answered that students or the committee decide the menu.
Patterns	A complaint sits until a larger incident forces action. When the Mess Office wants a quick change it tends to act alone. Complaints reach the operator, who points back to whoever owns the rule.
Structures	No confirmed rule requires committee approval before an operational change. Menu, kitchen execution, calendar and billing sit with different owners. Operators see which items waste (uttapam, set dosa and upma at Kadamba) and which move (puri, bonda, vada, idli) and do not choose the menu; the Bakul manager asks for a say, and his objection is to pairings, not dishes (kachori belongs with chole), untested with Bakul residents. Oversight of Kadamba runs CFS, CDS, Prism, operations manager, manager, supervisor, service staff, and covers safety and taste closely (daily tasting, weekly inspection, daily water reading) while no check asks how much was cooked against how many ate. The operator and an earlier institutional interview name different people as head of CDS, and the facility team's identity is also open. Complaints travel to the kitchen, never to the rule owner: a director's email to the manager and a WhatsApp group at Kadamba, the central dining team to the manager, kitchen and managing director at Bakul. Three student channels (in-app rating, email, Ping) have no confirmed case of changing registration, billing or hours. Change happens where the measurer also holds the lever: Yuktāhār is flat, with two project managers and peer cross-checking, and Kadamba cut ragi idli and raised plain idli from what it saw at the counter.
Mental models	The Mess Office appears to see these decisions as its own, with consultation optional under time pressure, and its stated reason for the December tightening was workload. At Yuktāhār waste is something to learn from, not punish, and staff treat each other ”like family”.

Root cause. Nobody owns the whole picture, and at the level of registration, billing and menu the system reacts only when something is loud. Oversight follows the same logic: intense on safety and taste, where failure is loud, and absent on quantity, where surplus is quiet. The kitchen mirrors it. At Bakul a run-out has a route to the managing

director and a surplus has none, and the operator's rule is to avoid the first at the cost of the second (candidate reading as the kitchen's answer to loudness). Inside a kitchen quiet numbers already work, because the person measuring also decides.

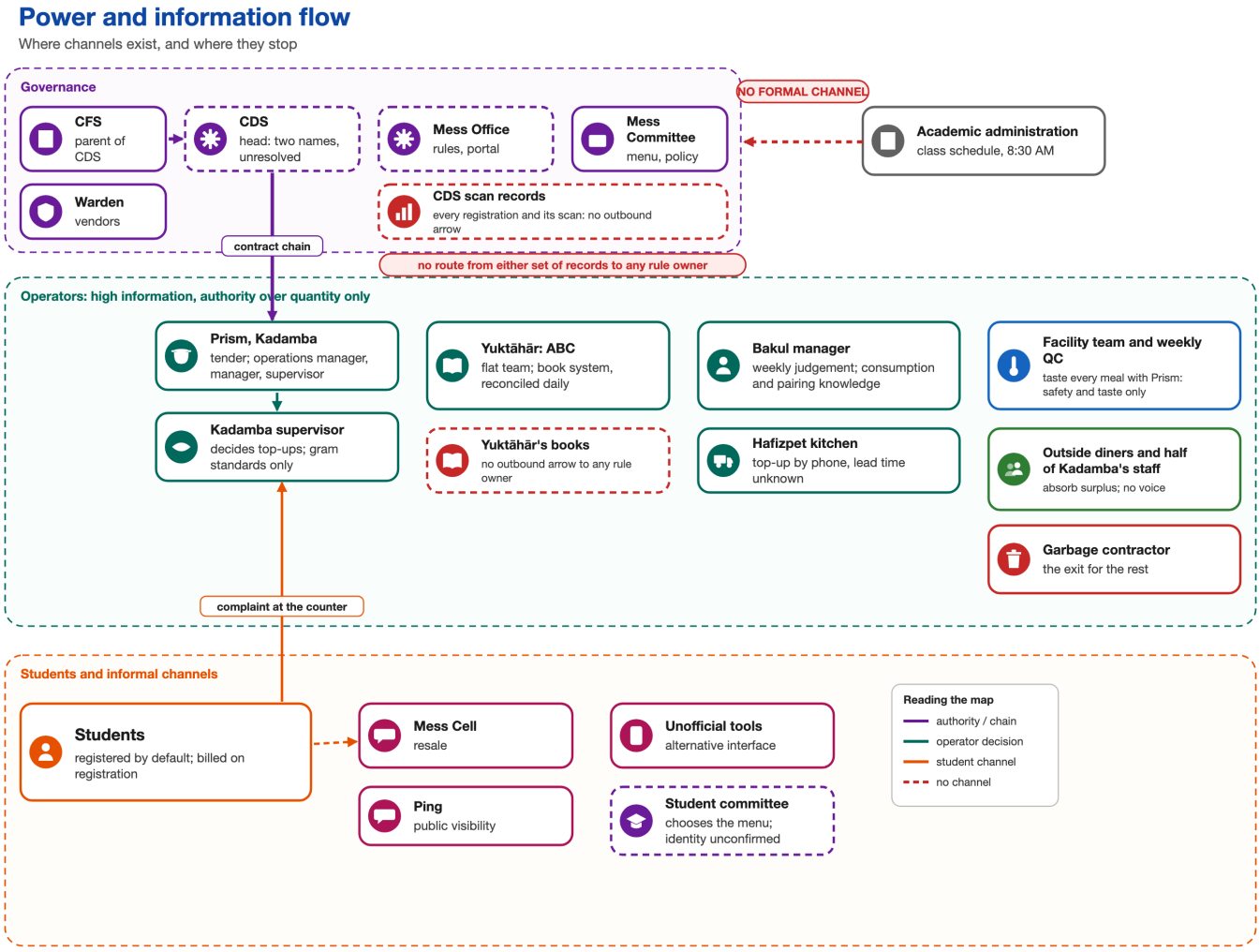


Figure 4. Power and information flow: where channels exist and where they stop.

Chain 4: A working fix already exists and gets switched off anyway

Layer	What the evidence shows
Events	During a gas shortage the mess cut to one item at the same price and the cancellation cap doubled from five to ten a month; Kadamba says it cut quantities and bought some items outside (single-sourced). On holidays and long weekends, Felicity being the named example, registration stopped being required: students could walk in and scan, or register a day or two ahead at any of the four messes, and anyone who neither registered nor came was not billed.
Patterns	By the CFS Chair's account it worked every time, and every time it was switched back off when the shortage or holiday ended. There is no sign of any review of keeping part of it.
Structures	The kitchen's own horizon for anything perishable is one day. Quantities are fixed in a meeting the day before at Yuktāhār, and Kadamba cooks a fixed share and tops up. Kadamba also cooks about 30 extra portions for 30 to 50 people outside the registration, such as night-shift security; an earlier noisy figure of 30 to 40 extra portions for paying walk-ins may be the same allocation, unconfirmed. Nothing exists to judge whether a temporary fix should become standing, and the cancellation cap is a number the institution has already shown it will move.
Mental models	Someone in registration and billing values an automatic, pre-committed count over matching food to demand, as a default and not a case-by-case judgement (candidate). Administrative workload is the likeliest source. Attendance-based billing seems acceptable only as an emergency measure.

Root cause. The counterexample to chain one: the institution has solved the student half twice and chose not to keep it. The kitchens do not need the four-day close. Several reasons remain unchecked. The per-plate payment may count registrations, so a walk-in model would cost the operator revenue (section 2 sets out the contested evidence). An automatic default is less administrative work. Or the relaxed model simply reverts with the calendar and nobody decides anything.

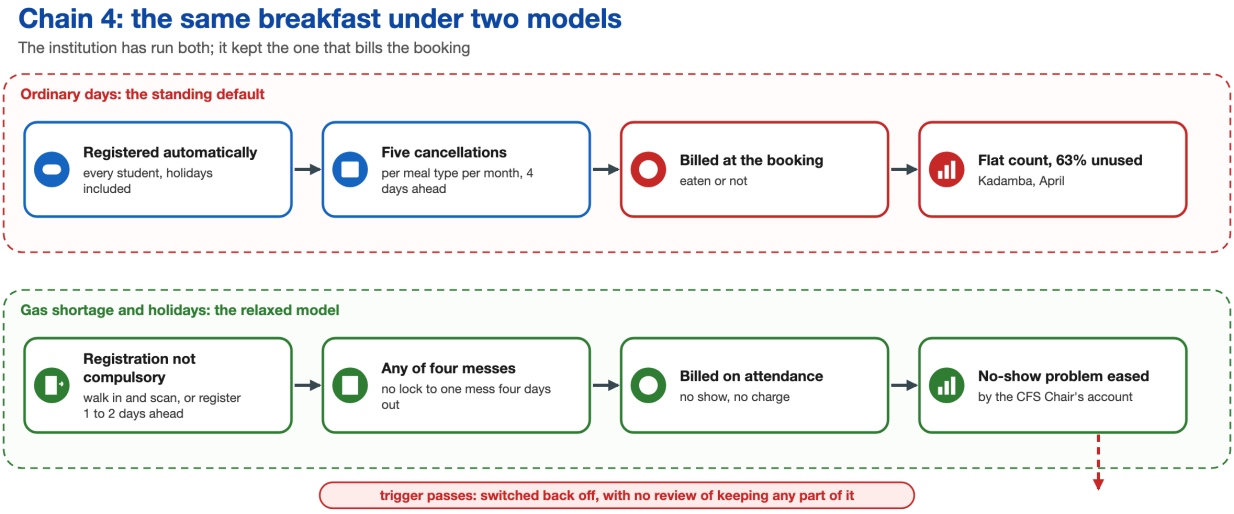


Figure 5. The same breakfast under the standing default and under the relaxed shock model.

Chain 5: A real fix, and a shadow it cast

Layer	What the evidence shows
Events	CDS convinced the Student Council, which convinced the Student Parliament, to replace one semester-long menu with a fortnightly rotation. Unprompted, the CFS Chair worried that a menu known in advance lets students decide not to come on days they dislike. On Thursday 3 September Yuktāhār planned breakfast from a sheet headed ”Thu wk 1&3”: the Thursday of weeks one and three share a menu and a sheet (confirmed).
Patterns	Menu-fatigue complaints dropped after the rotation, by the CFS Chair's account. Turnout moves with the item: non-veg registrations at Kadamba were used at 48.5% against 32.4% for veg (confirmed), biryani days reach about 700 registrations, and some items waste consistently while others run out. Bakul's manager names pav bhaji as a draw and muesli and Chocos as items that go. Whether students skip known bad-menu days has not been checked. Where a kitchen expects full turnout it drops its discount: Kadamba cooks biryani lunch at 100% of registrations, two pieces of 150 to 200 g with unlimited rice, and when some diners take a third piece later ones run short (lunch, single-sourced).
Structures	A real path runs from student complaint to policy: Student Council, Parliament, Committee approval, and this time it worked. Feedback is sorted by loudness, with low-intensity items going to the CFS head and high-intensity ones escalating. The menu is posted two weeks ahead and also changes monthly and seasonally, and the two-week cycle is what lets Yuktāhār plan each meal from the matching day in its Wastage Book.
Mental models	CDS believed menu variety was worth a policy change and acted on it, and is now questioning its own earlier belief that more menu transparency is automatically good. In the Bakul manager's account students choose combinations for balance, protein and preference while he reasons from traditional pairings and what is eaten, after about 20 years with North Indian food; that is his account of their reasoning, and whether familiar pairings matter to residents is untested.

Root cause. The opposite of chain three: proof that the system responds when a problem is loud enough. The new risk comes from the same structure that made the fix possible, since visibility that cures fatigue also enables avoidance. The rotation also gave kitchens a repeating unit to calibrate against, which a menu change breaking the two-week cycle would remove.

Chain 6: A meaningful slice of this isn't a problem at all

Layer	What the evidence shows
Events	One student fasts on purpose and stopped cooking breakfast years ago. Another says a friend's absence would not change whether they go, and a third says friends are a reason to go but never the deciding one.
Patterns	Late sleep is the reason given most often, ahead of class timing or food quality, and the records agree at pattern level: Sunday is the lowest day in both halls (27.3% at Kadamba, 17.9 to 24.1% on Yuktāhār's three legible Sundays) and has the latest arrivals (confirmed). The Yuktāhār site head adds that hackathons and exams push numbers down. Bakul's manager describes weekends as very low and gives two reasons, students not waking and students disliking the taste or menu; the first matches what students say and the second is his own hypothesis, since the six students interviewed rarely name food quality and none is known to live in Bakul. Legitimate variation has a ceiling: Kadamba's daily turnout ranged from about 25 to 48% and never neared lunch levels, so weekday and weekend variation cannot explain a 63% no-show rate (candidate).
Structures	Breakfast draws less company than lunch and sits outside a student's daily path. Only Yuktāhār offers a Jain line and counts it separately, about eight to fifteen Jain registrations per breakfast in September. The system registers everyone by default, so a legitimate absence, a fast or a trip, is counted as a no-show.
Mental models	For at least one student sleep outranks a booked breakfast whenever the two clash. The Bakul manager believes students skip because they may not wake or may not like the taste. A meal you have stopped cooking stops feeling like something you miss. The system assumes everyone should default into breakfast.

Root cause. This slice sits outside the billing and governance story. Some of what looks like the paradox is how a varied student body eats, and treating all of it as dysfunction would be a mistake. It is also too small to explain the gap.

Chain 7: The kitchen cooks to a guess, then absorbs the difference

Layer	What the evidence shows
Events	Shown the 18.7% Sunday row, the Yuktāhār site head said the kitchen would adjust ”not even next month, next week” (confirmed). Shown his own register, he still called breakfast ”about 50%”. On one biryani day 25 students arrived after close and the door was reopened (single-sourced; breakfast or lunch is unclear).
Patterns	The fixed ratio sits well above recorded turnout. Kadamba's 70% first batch against 37% actual is about 1.9 times what is eaten before any top-up. At Bakul the 70 to 80% share and the weekend floor of about 70% both exceed the highest turnout the manager himself expects, about 60%, so over-preparation is built in on his own numbers (candidate, no record to size it). Surplus leaves by several routes, none counted as institutional waste. Yuktāhār reuses it: leftover idli becomes idli upma or idli poha for lunch, dosa batter goes back to the fridge and surplus milk becomes curd (confirmed). At Kadamba about 30 extra portions feed 30 to 50 outsiders and about half the operator's own staff eat from the same food (single-sourced); whether the outsiders appear in any record the institution sees is unknown. At Bakul some leftovers return to Hafizpet and the rest is thrown. The rest goes to the bin. At Kadamba the team's observer judged that food is cooked ahead, reheated and moved between vessels until it degrades (candidate), so over-preparation costs quality as well as waste; at Bakul the equivalent cost is softening in transit. Waste is measured daily inside the kitchen: Yuktāhār weighs plate and kitchen waste per meal and logs percentage waste per item, and Kadamba reports about 5 to 10% daily with no stated base. In April Kadamba averaged about 1,072 registrations and 398 scans a day, so a 70% batch is about 750 portions and a surplus near 350 before any top-up (candidate arithmetic). A 5 to 10% waste figure fits that only if outsiders and staff, reheating, smaller top-ups or a different measurement base absorb most of it, and which is not known.
Structures	Yuktāhār plans at a twice-daily meeting from portal registrations (regular plus Jain plus cash meals), the Wastage Book entry for the matching day, the weekday attendance register, the month's pattern, and events, exams and holidays only if someone raises them (”We don't come to know until we sit in the meeting”). The Wastage Book works as a rule: ”If it runs out at 1:30 we re-cook; if at 2:20 we don't.” It runs a full set of paper books (Wastage Book, attendance register, wastage board, grocery issue book, store book, stock register, vegetable book and indent, utilities register, fridge logs, confirmed), reconciled by a two-book rule that the store's issues must match the kitchen's cooking, and a mismatch goes to the daily meeting. Kadamba sets 70% of registrations, veg and non-veg separately, with gram standards and a supervisor calling top-ups, and keeps no learning record. Bakul's manager lowers a share after surplus only down to his floor, and whether the weekly review rests on anything written is unknown. Operator records stay in the kitchen: nothing routes the Wastage Book or attendance register to whoever sets registration, billing or menu rules, and nothing routes the central scan records back to the kitchen. Kadamba's waste goes to a dustbin collected by a garbage contractor, with no composting.
Mental models	”Breakfast is about 50%” at the best-recorded mess, and ”at most about 60%” at Bakul with no record to test it. Three stances under one booking rule: Kadamba cooks to registrations as the job it is paid for, Yuktāhār treats waste as money and spends part of its margin on rock salt, cold-pressed oil and mill visits, and Bakul treats a shortage as the failure to avoid and waste as its accepted price. ”We don't have that idea” at Kadamba. At Yuktāhār, ”Prediction will be based on our data only... I want to remain close to the data”: it tried a commercial ERP and dropped it, since paying lakhs of rupees for prediction made no sense when the data sits in its books and its staff are not digitally inclined.

Root cause. Once cooked, the conversion from booking to plate cannot be reversed, and no kitchen can change the

number it starts from. In the two halls with records the breakfast gap is about as large under opposite philosophies, because the cause sits in the booking: calibration cuts waste and run-outs, not no-shows. The kitchen measures the gap in more detail than anyone and controls only quantities.

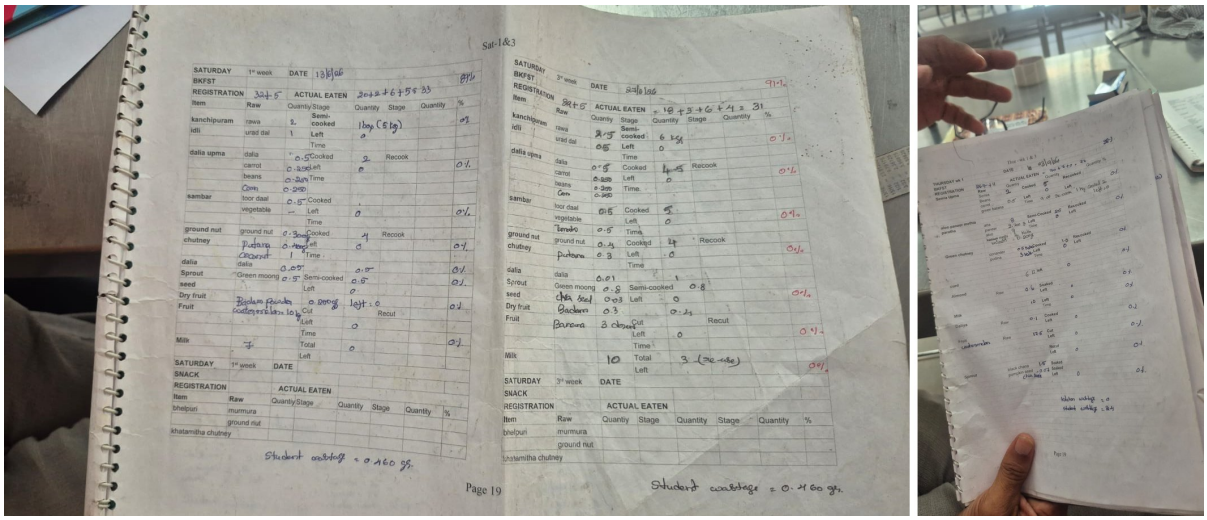


Figure 6. Yuktāhār's production and waste sheets for breakfast: issued, cooked, left, run-out time, re-cook and plate waste.

2. Structural Causes: What Keeps Repeating and Why

Across the seven chains three structural causes recur, and the problem splits into two halves: students not showing up for what they booked, and kitchens cooking more than is eaten.

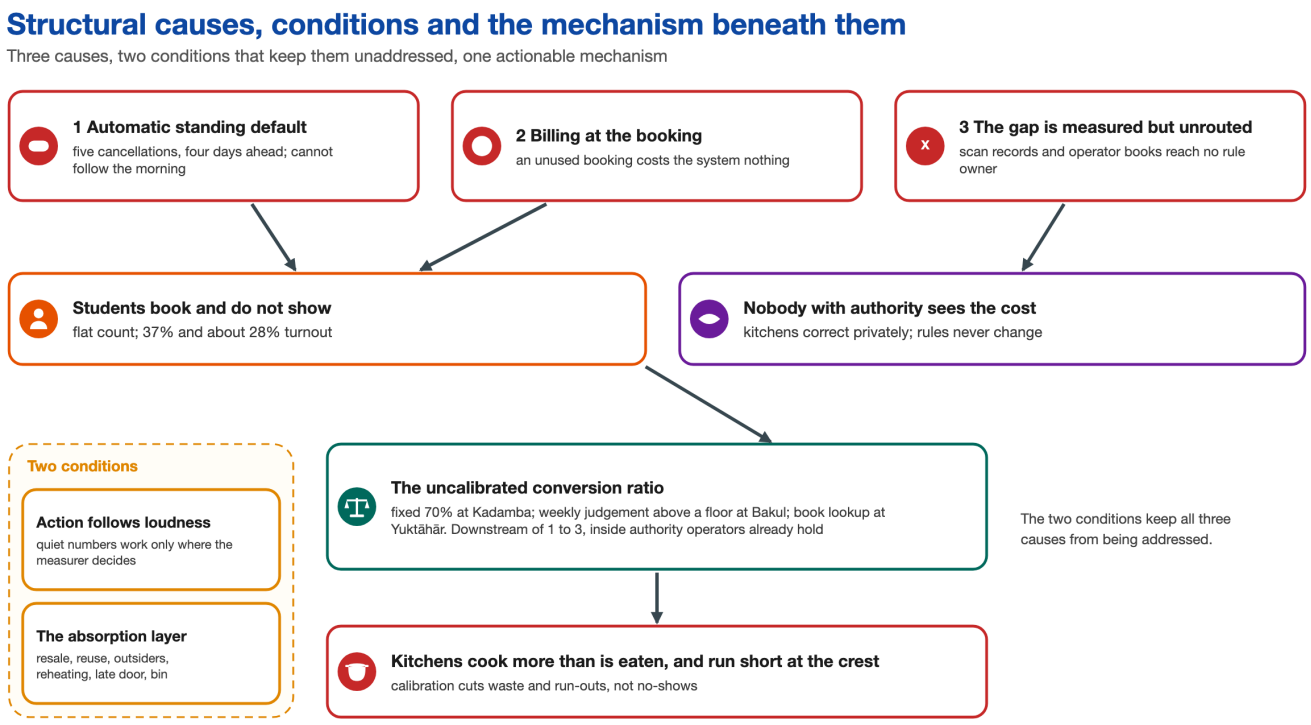


Figure 7. The structural causes, the two conditions that keep them unaddressed, and the conversion ratio beneath them.

Registration is an automatic standing default. With a narrow, early exit it cannot follow the decision each student

makes on the morning, and the records show its signature directly in the flat count. It also makes the problem breakfast's. Lunch runs under the same booking and mostly works, because the lunch decision is steadier.

Billing fires at the booking, not at consumption. This single rule explains the incentive to book broadly, why Skip Meal goes unused, why a cancellation cap is needed and why Mess Cell exists. It has been tested: billing followed attendance during the shortage and the holidays, and the no-show problem eased by the CFS Chair's account. It does not explain why kitchens cook above turnout.

The gap is measured but unrouted. Menu, kitchen quantity, calendar, registration and billing each sit with a different owner. The central system holds a timestamped scan for every registration, Yuktāhār holds per-item turnout and waste, and neither reaches a rule owner. The institution shows no sign of knowing the operator holds this data, and oversight of the kitchen checks safety and taste every day and never quantity. That is also why a fix that worked twice was never evaluated for keeping.

Two conditions keep all three unaddressed. **Action follows loudness at the institution level, while inside the kitchen quiet numbers already work.** Menu fatigue was loud and got fixed. The much larger registration gap is quiet and never travelled anywhere. At Yuktāhār the opposite holds: its own records cut waste between August and December with no complaint involved, because the measurer also decides and the measuring was made blame-free. The same logic shows on the operator side, where a Bakul shortage travels up to the managing director and a surplus goes nowhere. The **absorption layer** is the second condition. Resale and informal transfer, the operator's discount and live top-up, leftover reuse, outside diners and operator staff, held and reheated food (candidate), the walk-in buffer, late and back-door service, outside venues and finally the bin each take up a consequence of the gap before it becomes a signal anyone with authority sees.

Beneath the kitchen half sits one mechanism, **the uncalibrated conversion ratio**, which runs in three regimes.

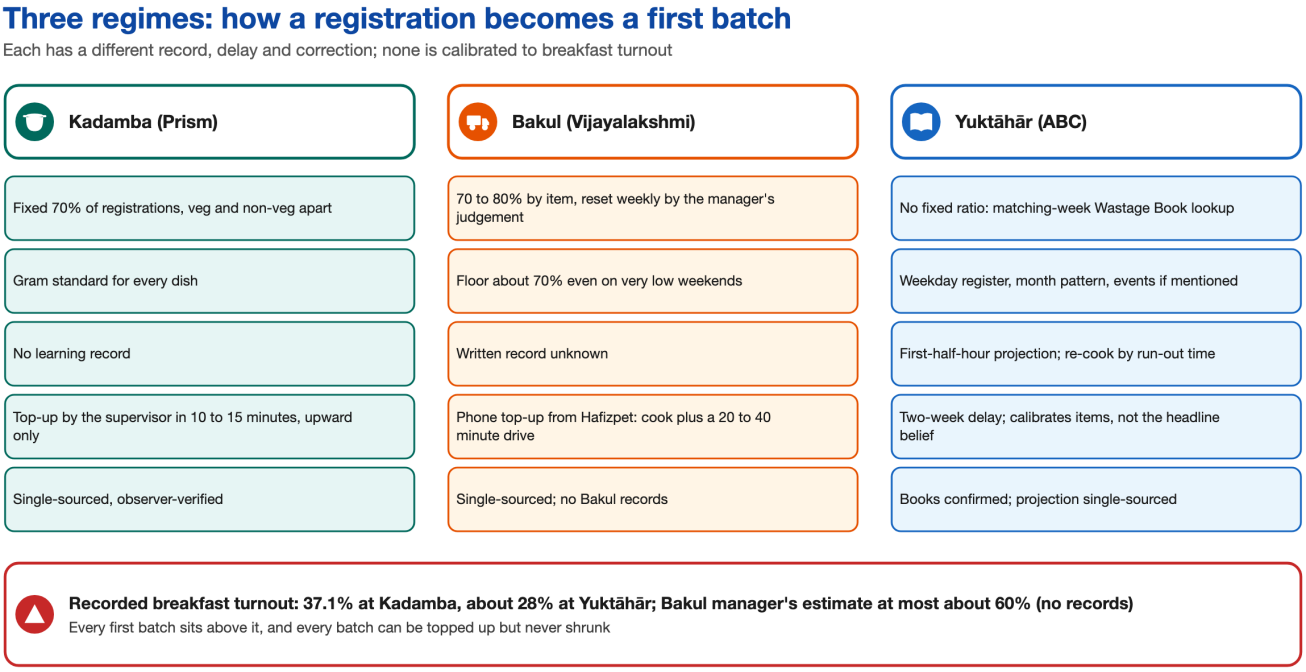


Figure 8. The three regimes that turn a registration into a first batch.

It is downstream of the booking and of the unrouted measurement, so it is not a root cause. It is the most actionable structure in the evidence because it sits inside authority operators already hold, one already calibrates and a second already adjusts weekly.

Each operator is paid monthly as a lump sum on plates, with deductions, and what a plate counts is contested. The Bakul manager says payment is on plates served (single-sourced, from one remark in the synthesis), while the

Kadamba operator's remark that it is paid for the registrations points the other way (candidate, not a contract term). Contracts may differ, one remark may not describe the formula, and no contract has been seen, so it is unknown for any operator whether turnout reaches its revenue. If Bakul is paid on served plates it bears the cost of its own over-cooking and accepts it, which fits a judgement that a shortage costs more than waste (candidate).

3. Reinforcing and Balancing Feedback Loops

Eighteen mechanisms were checked, numbered L1 to L17 plus L3b, the scheme used in the causal loop analysis. A loop counts only if the last step feeds back into the first and is evidenced. Three close, all balancing. Five close only through an assumed last step and are held as candidates. The rest stop partway and calling them loops would mislead.

Balancing loops that close. L4 is menu fatigue and rotation: fatigue drove feedback, feedback escalated through the Student Council, the Committee approved a rotation and variety rose, which lowers fatigue again; every step is confirmed by the CFS Chair and the operators' records. L11 is matching-day calibration at Yuktāhār: the Wastage Book entry for the matching day, with the month's pattern and events, sets the next quantity and turnout estimate, which produces a new leftover to record. Staff first feared blame when weighing began in August 2025, the managers made it blame-free, and the book was still filled in daily in September 2026. It calibrates item quantities, and the site head's headline figure stays above his register, so the headline belief does not drive planning (candidate). L11 is absent at Kadamba, which keeps gram standards and no learning record, and Bakul is a candidate second instance, bounded by its floor. L12 is in-meal top-up: the counter runs low, someone checks what is left against the time, more is cooked. It is confirmed at Yuktāhār, single-sourced at Kadamba and at Bakul, where the phone call and drive make it the clearest balancing loop with a delay. Its weakness is that it only moves upward and lags a crest.

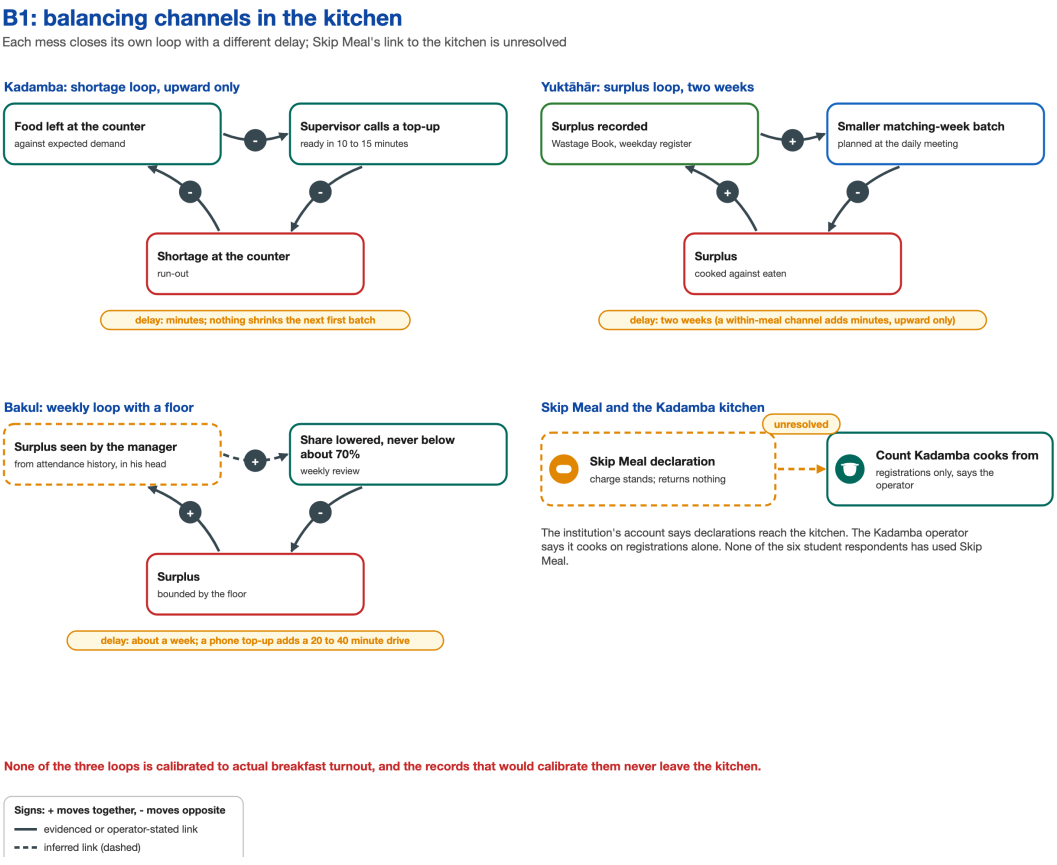


Figure 9. B1: how the three kitchens close their loops, with their different delays.

Candidate loops with an assumed last step. L1, loose registration and resale: registering loosely feeds resale, resale softens the cost of an unused registration, and that probably makes loose registration comfortable again;

most links are confirmed by direct quotes and the last is assumed, and the automatic default now explains flat registration without it. L3b, Kadamba scarcity: popularity drives defensive registration, which keeps it scarce on the platform and reinforces popularity, though at breakfast the scarcity is on the platform and not in the seats, with 1,070 registrations a day against a capacity of 1,200 and only 37% used. L9, surplus absorbed out of sight: reuse, outside diners and staff, held food, late service and the bin keep surplus from becoming pressure on whoever sets registration rules, and the closing step is assumed. L15, late-door tolerance: serving latecomers may teach students that arriving late is fine, which would keep the crest late; the first link is well evidenced and the closing link is assumed. L17, kitchen competence shifting the burden: the operators' local clearing of each morning's consequence plausibly stops any signal accumulating upstream, observed in the first step and assumed in the second.

Chains that look like loops and are not. L13, standing booking to over-preparation, is the core chain of the paradox: an automatic default keeps the count flat, Kadamba cooks a fixed share, over-preparation is worst where turnout is lowest, and nothing runs from the surplus back to the booking. L14, raising the ratio after a run-out, has only its first link evidenced, run-outs at the Kadamba crest, and Bakul's no-shortage rule supplies the premise as a stance and not an observed drift. A chain from surplus to reheated food to quality to skipping has no evidence for its last link, and Bakul has a distance counterpart in softening food. L6, the shock exception, is a calendar switch with no evidenced step feeding it back into its own reversal. L2, Skip Meal, runs one way and almost nobody uses it. L5, menu transparency, is a side effect of chain five. L7, late nights to skipping, has a confirmed first link and an untested rest. L8, vendor discipline, is unresolved until the plate basis is known. L16, operator waste to menu choice, is absent. L3, redistribution to Bakul or Palash, is contradicted: people lock down Kadamba harder when it is scarce. L10 is a stable stuck state and not a spiral.

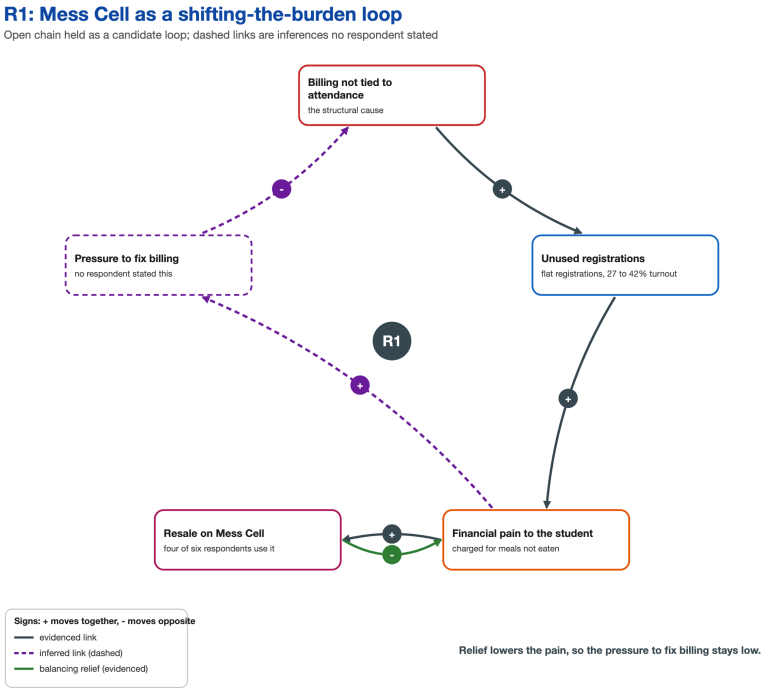


Figure 10. L1: Mess Cell as a shifting-the-burden loop; dashed links are inferences no respondent stated.

4. System Archetypes

Eleven standard archetypes were checked against the system.

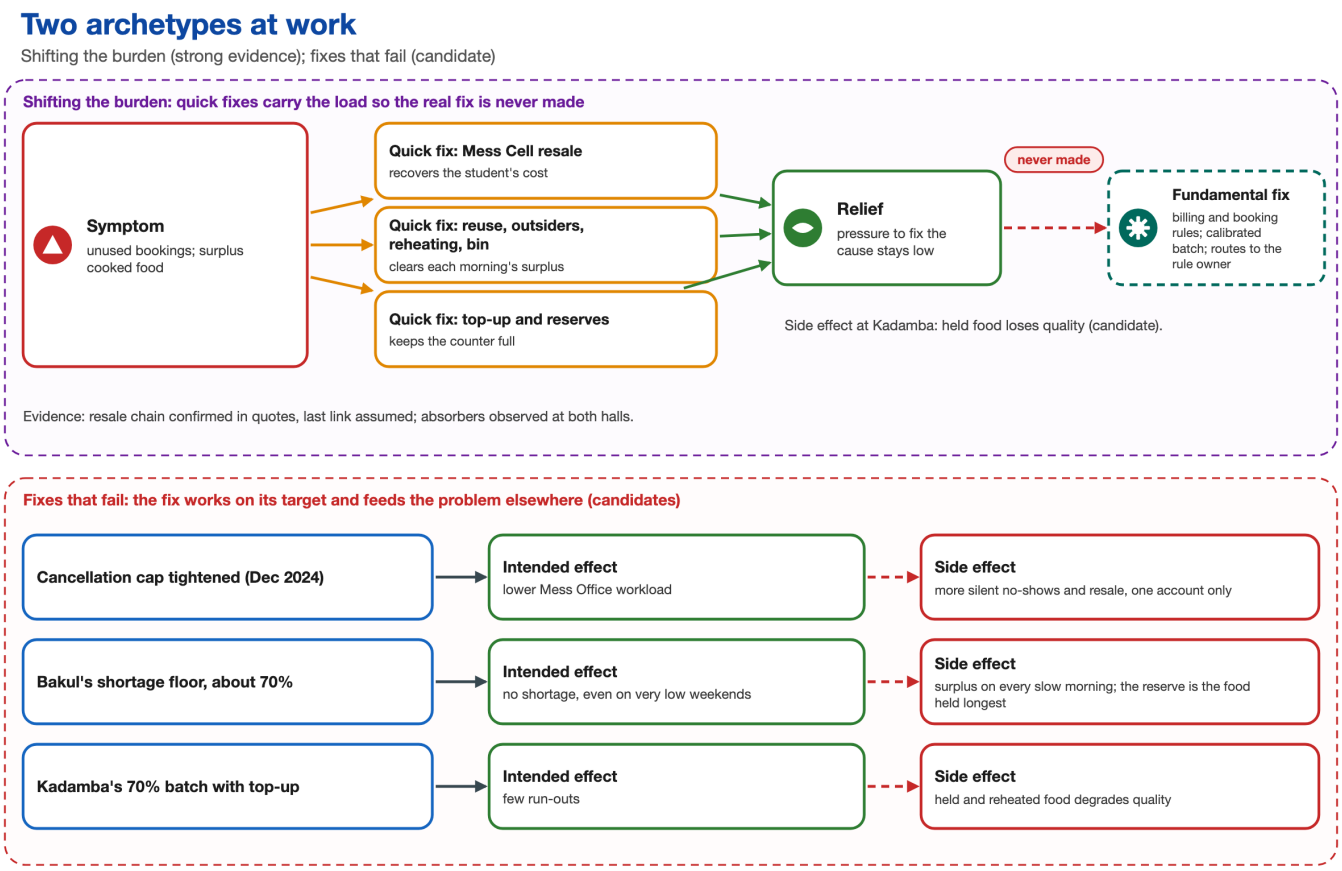


Figure 11. Two archetypes at work: shifting the burden, and fixes that fail.

Five hold up. **A quick fix standing in for a real one** appears twice: Mess Cell resale stands in for fixing billing, and reuse, outside diners and staff, held food and the bin stand in for calibrating the first batch. At Kadamba the substitute may carry a side effect, food held and reheated through service losing quality (candidate). **Growth hitting a ceiling** fits Kadamba's popularity, with Hafizpet's kitchen a candidate ceiling for Bakul and Palash. **Success to the successful** fits Kadamba's scarcity feeding defensive registration while Bakul stays underfilled. **A rule defeated by its own use** fits registering broadly to guard access: a Kadamba slot that is hard to get is used 37% of the time. Two more are real possibilities with too little evidence. **Fixes that fail** appears three times as a candidate: the December cap tightening may have worsened no-shows, though one account without before and after numbers supports it; Bakul's shortage floor solves shortage and feeds surplus on every slow morning; and Kadamba's 70% batch with top-up avoids run-outs and feeds reheated food. Four archetypes do not fit: access to meals is already limited by registration and a priced walk-in allowance, no two parties are shown escalating against each other, no stated target was quietly lowered, and every accidental-adversary candidate turned out to be one office's own trade-off.

5. Unintended Consequences

Billing on registration was meant to give the kitchen a stable number. It produced registrations out of habit with no cost to skipping, and a system that cannot tell a real no-show from someone defaulted into a meal. Automatic registration made the count flat, so the count carries nothing about the weekday pattern in who eats, and Kadamba does not rebuild it. The cancellation cap, meant to limit Mess Office workload, leaves students who exceed it or decide inside four days with no legitimate correction, which likely pushes them to silent no-shows or Mess Cell. Skip Meal, meant to cut preparation for a known no-show, gives the student nothing, so almost nobody uses it.

The fixed first batch was meant to stop kitchens cooking for every registration. It cooks below registrations and still over-prepares at breakfast, because 70% and 70 to 80% sit near Yuktāhār's lunch turnout and not the 28 to 37%

breakfast turnout. The menu rotation fixed boredom, created a new way to plan around a menu known to be disliked, and gave Yuktāhār a repeating unit to calibrate against. The shortage exception got the mess through a supply problem and proved, by accident, that the standing model can be replaced by one that tracks attendance, which no one has revisited. The food-safety sequence keeps the mess compliant and also locks the first batch in before any turnout signal (candidate). Waste does appear as a number at Yuktāhār, and the number stays in the kitchen. Any new measurement risks the blame reaction that weighing first met there. Outsourcing was meant to save money and modernise hygiene. It put the most detailed records with the party that has least say over registration, billing and menu, and produced three stances on waste.

6. Systemic Tensions

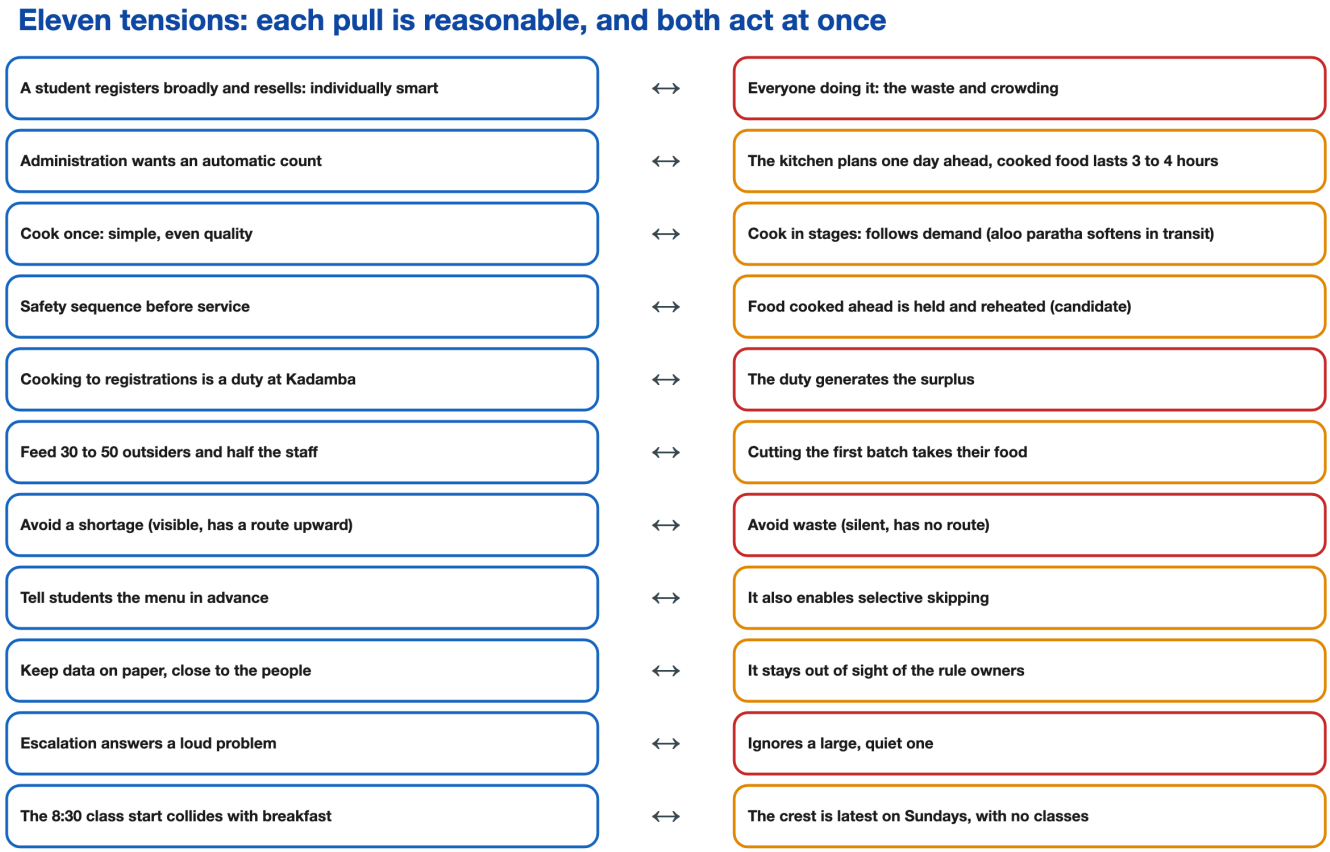


Figure 12. *Eleven tensions: each pull is reasonable, and both operate at once.*

A student registering broadly and reselling the rest is individually smart, and added up it is the waste and crowding the analysis describes. Registration administration wants an automatic count while the kitchen plans one day ahead with three to four hours of cooked life. Both the standing system and the shock exception have been run by the same institution, which shows the tension between a fixed count and real demand is real. Cooking once is simpler and keeps quality even, and cooking in stages follows demand; Yuktāhār cooks once because ”student fluctuation is very high”, and at Bakul aloo paratha started around 3 AM for a 7 AM arrival loses texture in the box. A safety sequence of sampling, display and tasting keeps food safe, and with cooking ahead it leaves food held and reheated through a two-hour service (candidate).

At Kadamba cooking to registrations is framed as duty, and the duty generates the surplus. At Yuktāhār waste reduction is a profit lever the operator spends partly on quality, so profit and quality point the same way. Staying close to the data on paper keeps planning owned by the people who do it, and keeps the data out of sight of the rule owners. Feeding outsiders and staff from the same food is a welfare benefit that absorbs surplus uncounted, and

any cut to the first batch takes their food unless they are planned for. Kadamba's own figures leave an unresolved tension: a 70% batch against 37% turnout implies a surplus near 350 portions a day, while the operator reports 5 to 10% waste on no stated base.

Avoiding a shortage and avoiding waste pull against each other at every first batch, and Bakul has settled it by rule: no shortage even if food is wasted. A shortage is visible and has a route to the managing director while a surplus is silent, which also explains the reserve at 9:20. The menu is chosen for nutrition and preference and, in the Bakul manager's account, eaten by habit and pairing (his view, untested). Telling students the menu in advance fixes boredom and enables selective skipping. A registration system built for one kind of student leaves little room for someone who fasts or eats differently, and folds that into the same no-show figure. Escalation answers a loud problem and ignores a large quiet one. Nobody has asked the academic side to move the 8:30 class start and this document does not suggest the timetable should change. The collision is real but should not be over-read, since the crest is latest on Sundays, when there are no classes (candidate).

7. Power Asymmetries

Students carry nearly all the visible cost: money spent on meals never eaten, crowding and running out. They control almost none of the rules that produce it, and they are registered by default with no way to nullify the charge.

The operator is data-rich and authority-poor. Neither Kadamba, which sets its own ratio, nor Yuktāhār, which holds per-item daily records, has any part in the registration rules that generate its starting number, or in the menu. The Bakul manager holds two kinds of knowledge the menu process does not use, which items are eaten and which pairings work, and his weekly calibration may live only in his head (that this knowledge would improve the menu is a candidate). Inside the operators power splits differently. At Kadamba the person who decides quantities in service is a supervisor several levels below the vendor and further still from CDS; at Yuktāhār the measurers and deciders are one small peer group. The central dining side holds a timestamped scan for every registration and nothing shows anyone analysing it. Students collectively, through Mess Cell, hold a real-time read on who is coming that counts formally nowhere.

The academic side holds total control over the schedule collision that appears in nearly every interview, and faces none of its consequences. When asked whether anyone had raised the 8:30 overlap with them, nobody really had. The calendar reaches Yuktāhār's kitchen only by word of mouth (confirmed). The Committee sets the menu without a view of what is eaten against what is registered, and the dropped dinner sambar reached students as an unexplained cut. The Warden holds authority over vendor contracts and structural change without a confirmed link to the numbers. The facility team and weekly inspection hold authority over safety and taste and none over quantity. CDS's chair, whose office sits directly above the vendor, has not been interviewed. Mess Cell runs with complete flexibility and no oversight, and a student's own registration behaviour (register, cancel, resell, or not show) costs the same either way. Yuktāhār counts faculty and staff diners separately, zero to two per breakfast (confirmed).

8. Workarounds: What Students and Operators Have Built Around the System

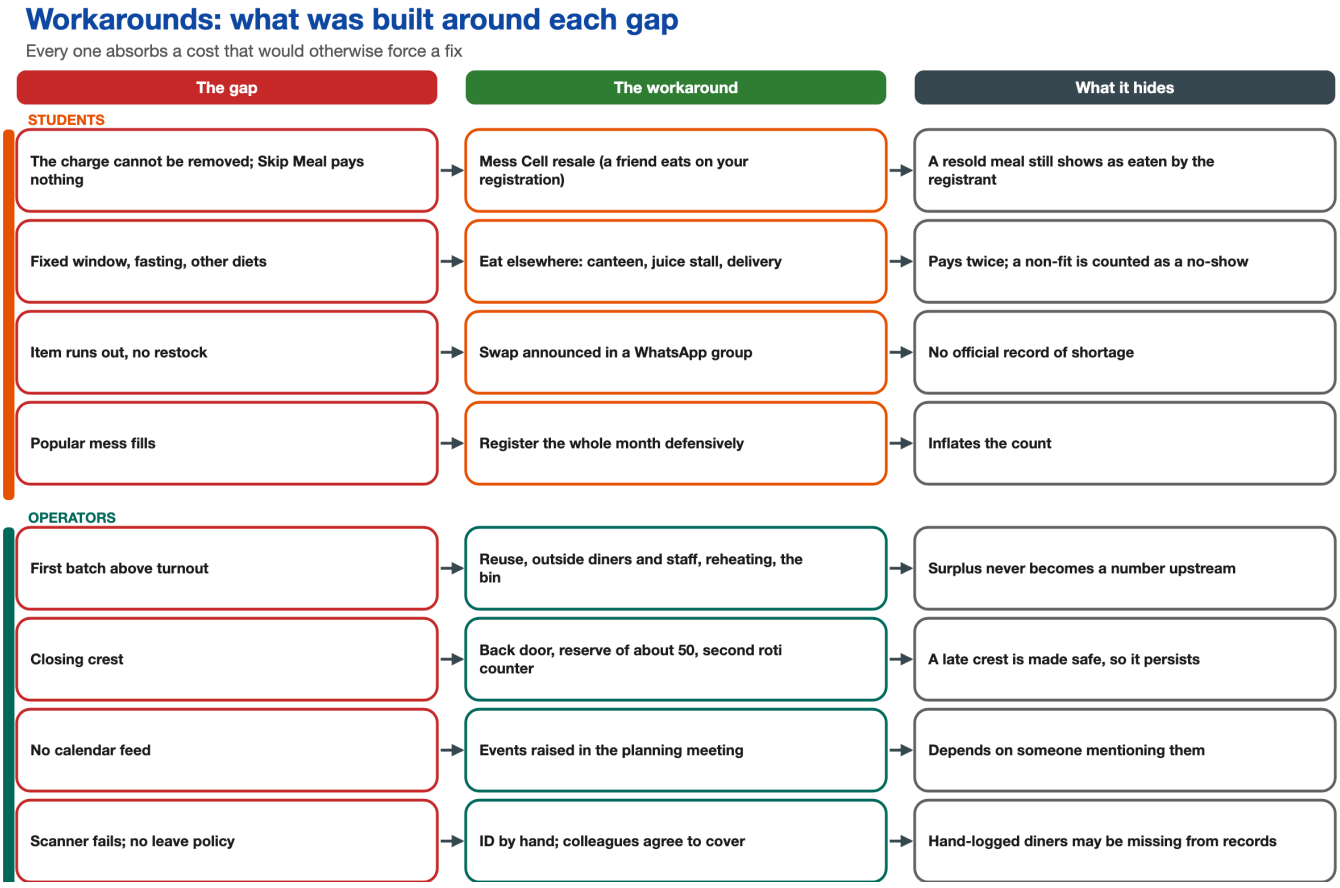


Figure 13. *Where the system leaves a gap, and what people have built to absorb it.*

Every workaround quietly absorbs a cost that would otherwise force someone to fix the real problem. Mess Cell absorbs the financial pain of over-registering, so the pressure to fix billing never builds, and it hides the gap, since a resold meal probably still appears as eaten by the original registrant; the scan export has no student ID, so the records cannot show it. Skip Meal was built to solve this, and students replaced it with Mess Cell because Mess Cell pays. Eating elsewhere (the canteen, a juice stall, a delivery app) gets around a fixed window and a fasting calendar, usually at the price of paying twice. When something runs out, staff post the swap in a WhatsApp group after the fact, which keeps trust without any official record. Students whose dietary or cultural needs the menu does not cover mostly just do not come.

Operators have built their own layer, described in chain 7 and drawn above. Beyond the absorbers already named, Yuktāhār holds raw material to an impurity norm, opens two roti counters at peak and covers absences by agreement in the daily meeting. Kadamba writes ID and name by hand when the scanner fails and bought items outside during the gas shortage. Bakul phones Hafizpet, holds its reserve and switched paneer suppliers twice after complaints. Kadamba does not compost, and where its contractor takes the waste is unknown, as is Yuktāhār's disposal route.

9. Synthesis

Breakfast registrations are used 37% of the time at Kadamba and about 28% at Yuktāhār, against 56 to 89% for lunch. Attendance is low partly because every student is registered by a default that cannot follow the morning, partly because registering costs nothing extra if you do not show and nothing removes the charge, partly because the gap is measured in two places and owned by no one, and partly because a real share is just how a varied student body eats. On the kitchen side Kadamba turns a flat default into a first batch at a fixed ratio above turnout, Bakul adjusts weekly by judgement and never below a shortage floor, and Yuktāhār plans from its books. Each tops up only upward and too late for the crest, and absorbs the rest by reuse, outsiders, reheating, late service and the bin.

The institution has already proven it can fix the student side. It found a way to charge people for what they eat, ran it more than once and did not keep it. The kitchens do not need the four-day close, so what holds the standing model in place is unknown: the per-plate payment, the convenience of an automatic default, or a reversion with the calendar. The trade-off has never been weighed on purpose, because no single actor holds both the data and the authority, and because the system attends to loud problems and not large, quiet ones.

Several things stand out. The fix exists and has worked more than once. The student half comes from an automatic standing booking with billing tied to it, both optional since the alternative was tested. The kitchen half comes from a conversion inside operators' own authority. The two best-recorded halls are a natural comparison, a fixed ratio with no learning record against a book-based system with blame-free measurement, and they produce about the same gap, so calibration reduces waste and run-outs and does not move turnout. A real share of the gap is not a problem to fix, though weekday variation alone cannot explain a 63% no-show rate.

Three questions decide what comes next. What does the per-plate payment count, registrations or scans? Why must Kadamba guess its ratio when the data to calibrate it is recorded and one operator already does? And where does Kadamba's surplus go, given 5 to 10% reported waste against a 70% batch at 37% turnout? Several source conflicts stay open until someone asks directly: whether Skip Meal changes what Kadamba cooks, whether the CDS head named by the operator and the one named earlier are two people or one, whether the Yuktāhār head's 50% is an average or a ceiling, whether Kadamba's outside diners appear in any record, how long a Hafizpet top-up takes, and whether Bakul's weekly review rests on a written record. Until someone asks, this will likely keep repeating, not because the answer is unknown, but because the records that hold it never reach the people who set the rules.